

Summary

Computational physicist with six years of production software experience in computer vision and deep learning, specialising in E(3)-equivariant machine learning for quantum materials simulation. Author of DSpinGNN, a physics-informed equivariant GNN for mesoscale spin-lattice dynamics in strain-engineered 2D magnets, submitted to Computational Materials Science (Elsevier).

Education

MS Physics

Lahore University of Management Sciences**Sept 2023 - July 2026**

Supervisor: Dr. Muhammad Sabieh Anwar

Co-supervisor: Dr. Muhammad Faryad

Thesis: DSpinGNN: A Physics-Informed Equivariant Graph Neural Network for Dynamic Magnetic Exchange Prediction in Strain-Deformed Monolayer CrI₃ (submitted, Computational Materials Science (Elsevier))

COURSEWORK:

Density Functional Theory – online course, Coursera (certificate) | Quantum Information Science & Technology A+ | Quantum Computing A+ | Quantum Algorithms A+ | Electrodynamics A | Statistical Mechanics A | Electrochemical Energy Storage & Conversion A

BS Computer Science

Lahore University of Management Sciences**Sept 2013 - June 2017**

Capstone: Visual Inertial Odometry – robust localization via Multi-State Constraint Extended Kalman Filter (MSCKF), fusing visual feeds with kinematic robot models

Research: Funded research internship, University of Siegen, Germany – computational experiments for spatial stereo setups

Relevant Coursework: Machine Learning, Applied Probability, Applied Statistics, Applied Stochastic Processes

Publications

- I. A. Balghari, M. Faryad, and M. S. Anwar, "DSpinGNN: A Physics-Informed Equivariant Graph Neural Network for Dynamic Magnetic Exchange Prediction in Strain-Deformed Monolayer CrI₃," Preprint at arXiv: arxiv.org/abs/2606.11685
- I. A. Balghari and M. S. Anwar, "PhysTrackX: An Open-Source Framework for Automated Tracking and Sensor Synchronization," Apparatus Note in final preparation for submission to the American Journal of Physics (2026).

Professional Skills

AI for Science:

PyTorch, e3nn, NequIP, Geometric Deep Learning E(3)-Equivariant Graph Neural Networks, e3nn (equivariant convolutions, tensor products, spherical harmonics)

Computational Physics:

Density Functional Theory (DFT+U), Quantum ESPRESSO, Wannier90, TB2J, SLURM, Statistical Mechanics, ASE, OVITO, FIRE relaxation

Software & Systems:

Python, C/C++, Git, Linux & Systems, Pipeline architecture, Compute optimisation

Sabieh Anwar Research Group, LUMS, Pakistan

Thesis project: DSpinGNN — Physics-Informed Equivariant GNN for Dynamic Magnetic Exchange Prediction in Strain-Deformed CrI_3 (submitted, Computational Materials Science (Elsevier))

- Developed DSpinGNN [\[code\]](#), a bifurcated $E(3)$ -equivariant architecture coupling a NequIP-inspired GNN (implemented using e3nn tensor product operations and PyTorch) for Langevin structural dynamics with a physics-informed Δ -MLP exchange predictor that embeds the Goodenough-Kanamori superexchange relationship as a closed-form analytical inductive bias — making predictions interpretable in terms of established quantum chemistry.
- Achieved test-set MAEs of 1.1 meV/atom (energy), 6.5 meV/Å (force), and 0.18 meV (exchange coupling, $R^2 = 0.91$) on a stratified 61-configuration held-out set, with all three quantities evaluated simultaneously against DFT+U ground truth.
- Demonstrated 400× system size transferability without retraining: deployed an 8-atom-trained model on a 3,200-atom supercell, driving stable Langevin dynamics at 5 K and extracting mesoscale observables inaccessible to direct DFT — domain wall width $\xi = 1.7 \pm 0.3$ nm and interference oscillation period $\tau = 0.27$ ps, consistent with experimentally verifiable mesoscale features.
- Built SpinDFT [\[code\]](#), an end-to-end HPC orchestrator automating ~ 470 DFT+U $\rightarrow$ Wannier90 $\rightarrow$ TB2J calculations with dynamic Fermi-level calibration, crash-proof NSCF diagonalization, and just-in-time wavefunction deletion; released as an independent open-source repository alongside DSpinGNN.
- Reproduced DTNN and NequIP using PyTorch + e3nn as foundational preparation, independently implementing equivariant message passing, tensor field operations, and force prediction heads without reference to the authors' codebases.

Independent Project: (Software Released)

- Scientific Software (with Dr. M. S. Anwar): Designed and shipped PhysTrackX, an open-source video-tracking and sensor-synchronization framework for experimental physics, now used in teaching laboratories (16 releases, AJP apparatus note in preparation). [\[code\]](#)

- Executed computational experiments for spatial stereo vision setups as part of a funded international research placement; gained hands-on experience with 3D reconstruction pipelines and multi-camera calibration.

Teaching Experience

Delivered tutorial sessions and office hours for junior students on visual feature extraction, epipolar geometry, and matrix operations underlying modern visual algorithms; supported course delivery for one semester during BS enrolment.

Delivered tutorial sessions and supported instruction of a graduate-level course integrating quantum information theory with modern machine learning; assisted students with problem sets spanning variational quantum circuits, quantum kernel methods, and hybrid classical-quantum architectures.

Awards And Achievements

Dean's Honour Roll

2026

Awarded for outstanding CGPA maintained during MS studies.

LUMS Merit Scholarship

2024-2025

Awarded for exceptional academic performance during MS Physics; competitive departmental scholarship.

Facebook Hacker Cup - Qualification Round

2016

Google Code Jam - Qualification Round

2017

Qualified for both elite international algorithmic programming competitions; top percentile performance in combinatorial optimisation and algorithm design under timed conditions.

Professional Software Engineering Experience

Senior Software Engineer

Roll Camera Inc, Remote

Feb 2021 - Feb 2025

[Concurrent with MS Physics, LUMS from Sep 2023]

- Overhauled the core deep learning inference engine, reducing total compute time by 60× through pipeline architecture restructuring and operator-level optimisation.
- Designed and deployed an AI-driven 3D visual effects system end-to-end, establishing deployment standards that accelerated enterprise product shipping by 70%. [\[Sample\]](#)
- Resolved critical architectural bottlenecks under strict production deadlines to stabilise high-demand features for early enterprise adoption.

Independent Engineer

Remote

Jan 2020 - Jan 2021

- Independently conceived and deployed an AI-driven spatial tracking pipeline for 3D vehicle detection and kinematic analysis in heavily occluded environments – self-directed research bridging industry CV work and formal academic inquiry. [\[Code\]](#) [\[Sample\]](#)

Software Engineer

Vidimensions, Singapore

Jan 2019 - Dec 2019

- Engineered a statistical ensemble fusion methodology combining multiple deep learning architectures for complex spatial analytics tasks.
- Improved anomalous behaviour detection accuracy by 10% through enhanced mathematical modelling of spatial decision boundaries.

Software Engineer

Hazen.ai, Pakistan

Aug 2017 - Dec 2018

- Designed and implemented non-linear Kalman filters for object kinematics modelling, improving trajectory tracking precision by 20%. [\[Sample\]](#)
- Led architectural development of a 3D vehicle detection prototype deploying spatial constraints for occlusion-robust performance.